



BobSim

SHARK Kinematic Overlay

2027 suspension kinematics against the Orion baseline

Generated: 2026-09-23 02:01

Run Notes

- Baseline: Orion (vehicle.yml). Variant: 2027.
- Ranking tolerances: deg: 0.05, mm: 0.5. A curve is significant when its peak change exceeds one tolerance.
- Bump 21 points over -40..+40 mm, roll 21 points over -1.50..+1.50 deg, steer 21 points over -30..+30 mm rack (front axle only). Zero is a sample point in every sweep.
- Datum gate closed: 10 z-dependent curves withheld.
- Full run notes are on the next page.

Run Notes

- SHARK file describes the REAR axle only; the front axle, and all mass/aero/tyre/powertrain data, are unchanged from the baseline.
- TODO(datum): Conflicting datum evidence. Reading the trailing scalar as the loaded rolling radius (199.890 mm), a shared ground-plane datum implies a contact patch at $z = 0$. The baseline lands at +0.008 mm (consistent); the SHARK wheel centre lands at +1.198 mm (inconsistent). TITLES carries no annotation to settle it. The +1.190 mm offset is therefore NOT confirmed as a ride-height change, and z-dependent curves are withheld until it is.
- rod_to changed 'upper' -> 'lower' (pushrod picks up on the other wishbone in the SHARK model)
- Anti-roll bar not imported (SHARK defines none). It is out of scope for kinematics and handled outside this tool.
- architecture.rear: bellcrank_stabar -> bellcrank
- Imported tests/fixtures/rear_pushrod.shk onto vehicle.yml -> vehicle_2027.yml
- DATUM GATE CLOSED - rear: datum unresolved. Z-dependent outputs withheld.
- Sweep ranges match the app registry defaults; the point count is raised to 21/21 so the design position is sampled rather than falling between two points as it does on the app's even grid.
- Steer sweep: 21 points over -30..+30 mm rack displacement, front axle only; plotted against the solved road-wheel steer angle rather than the commanded rack travel.
- The anti-roll bar takes no part in the kinematic solve; it is handled outside this tool.

Withheld curves (datum gate closed):

- bump_front_ic_z_mm
- bump_front_swing_arm_mm
- bump_rc_height_mm
- bump_rc_migration_mm
- bump_rc_z_mm
- roll_front_ic_z_mm
- roll_front_swing_arm_mm
- roll_rc_height_mm
- roll_rc_migration_mm
- roll_rc_z_mm

Summary - design position and working-range gradient (1/3)

Values at zero are read from the sweep, not extrapolated. Slope is the least-squares gradient over the full swept range. 'x tol' is how many engineering tolerances the peak change spans.

Curve	Axle	Orion @0	2027 @0	delta @0	Orion slope	2027 slope	peak delta	x tol
Bump Front-View IC y	rear	-1179.431 mm	-615.036 mm	564.395 mm	11.789 /mm	12.356 /mm	564.395 mm	1128.8
Roll Front-View IC y	rear	-1179.431 mm	-615.036 mm	564.395 mm	122.204 /deg	122.940 /deg	564.395 mm	1128.8
Roll RC y-Migration	rear	0.000 mm	0.000 mm	0.000 mm	265.919 /deg	85.447 /deg	270.967 mm	541.9
Bump Kingpin side-view inclination	rear	-0.130 deg	5.895 deg	6.026 deg	0.003 /mm	0.002 /mm	6.221 deg	124.4
Roll Kingpin side-view inclination	rear	-0.130 deg	5.895 deg	6.026 deg	0.035 /deg	0.024 /deg	6.067 deg	121.3
Bump KPI	rear	15.586 deg	15.782 deg	0.196 deg	0.034 /mm	0.048 /mm	1.046 deg	20.9
Bump Camber	rear	0.000 deg	0.000 deg	0.000 deg	-0.034 /mm	-0.048 /mm	0.847 deg	16.9
Roll KPI	rear	15.586 deg	15.782 deg	0.196 deg	0.361 /deg	0.507 /deg	0.460 deg	9.2
Roll Scrub Radius	rear	-4.133 mm	-1.264 mm	2.869 mm	-0.004 /deg	-0.003 /deg	2.869 mm	5.7
Bump Scrub Radius	rear	-4.133 mm	-1.264 mm	2.869 mm	-0.000 /mm	-0.000 /mm	2.869 mm	5.7
Roll Camber	rear	0.000 deg	0.000 deg	0.000 deg	-0.361 /deg	-0.506 /deg	0.263 deg	5.3
Bump Mechanical Trail	rear	-5.355 mm	-2.894 mm	2.460 mm	-0.000 /mm	0.000 /mm	2.478 mm	5.0
Roll Mechanical Trail	rear	-5.355 mm	-2.894 mm	2.460 mm	-0.002 /deg	0.001 /deg	2.465 mm	4.9
Bump Toe	rear	-0.000 deg	0.000 deg	0.000 deg	0.002 /mm	-0.000 /mm	0.180 deg	3.6
Roll Toe	rear	-0.000 deg	0.000 deg	0.000 deg	0.020 /deg	-0.000 /deg	0.046 deg	0.9
Bump Camber	front	0.000 deg	0.000 deg	0.000 deg	-0.026 /mm	-0.026 /mm	0.000 deg	0.0
Bump Toe	front	0.000 deg	0.000 deg	0.000 deg	-0.000 /mm	-0.000 /mm	0.000 deg	0.0
Bump Caster	front	4.564 deg	4.564 deg	0.000 deg	-0.002 /mm	-0.002 /mm	0.000 deg	0.0
Bump KPI	front	7.758 deg	7.758 deg	0.000 deg	0.026 /mm	0.026 /mm	0.000 deg	0.0
Bump Mechanical Trail	front	12.465 mm	12.465 mm	0.000 mm	-0.001 /mm	-0.001 /mm	0.000 mm	0.0
Bump Scrub Radius	front	26.963 mm	26.963 mm	0.000 mm	0.002 /mm	0.002 /mm	0.000 mm	0.0
Bump RC y-Migration	front	0.000 mm	0.000 mm	0.000 mm	0.000 /mm	0.000 /mm	0.000 mm	0.0
Bump RC y-Migration	rear	0.000 mm	0.000 mm	0.000 mm	0.000 /mm	0.000 /mm	0.000 mm	0.0
Roll Camber	front	0.000 deg	0.000 deg	0.000 deg	-0.276 /deg	-0.276 /deg	0.000 deg	0.0
Roll Toe	front	0.000 deg	0.000 deg	0.000 deg	-0.001 /deg	-0.001 /deg	0.000 deg	0.0
Roll Caster	front	4.564 deg	4.564 deg	0.000 deg	-0.024 /deg	-0.024 /deg	0.000 deg	0.0

Summary - design position and working-range gradient (2/3)

Values at zero are read from the sweep, not extrapolated. Slope is the least-squares gradient over the full swept range. 'x tol' is how many engineering tolerances the peak change spans.

Curve	Axle	Orion @0	2027 @0	delta @0	Orion slope	2027 slope	peak delta	x tol
Roll KPI	front	7.758 deg	7.758 deg	0.000 deg	0.275 /deg	0.275 /deg	0.000 deg	0.0
Roll Mechanical Trail	front	12.465 mm	12.465 mm	0.000 mm	-0.011 /deg	-0.011 /deg	0.000 mm	0.0
Roll Scrub Radius	front	26.963 mm	26.963 mm	0.000 mm	0.019 /deg	0.019 /deg	0.000 mm	0.0
Roll RC y-Migration	front	0.000 mm	0.000 mm	0.000 mm	144.903 /deg	144.903 /deg	0.000 mm	0.0
Bump Front-View IC y	front	-1581.786 mm	-1581.786 mm	0.000 mm	8.752 /mm	8.752 /mm	0.000 mm	0.0
Roll Front-View IC y	front	-1581.786 mm	-1581.786 mm	0.000 mm	92.467 /deg	92.467 /deg	0.000 mm	0.0
Steer Camber	front	0.000 deg	0.000 deg	0.000 deg	0.079 /deg	0.079 /deg	0.000 deg	0.0
Steer Scrub Radius	front	26.963 mm	26.963 mm	0.000 mm	-0.203 /deg	-0.203 /deg	0.000 mm	0.0
Steer Mechanical Trail	front	12.465 mm	12.465 mm	0.000 mm	0.460 /deg	0.460 /deg	0.000 mm	0.0
Steer RC y-Migration	front	0.000 mm	0.000 mm	0.000 mm	0.000 /deg	0.000 /deg	0.000 mm	0.0
Steer RC z-Migration	front	-31.724 mm	-31.724 mm	0.000 mm	-0.077 /deg	-0.077 /deg	0.000 mm	0.0
Steer KPI	front	7.758 deg	7.758 deg	0.000 deg	0.002 /deg	0.002 /deg	0.000 deg	0.0
Steer Caster	front	4.564 deg	4.564 deg	0.000 deg	-0.000 /deg	-0.000 /deg	0.000 deg	0.0
Bump Front-View Swing Arm	rear	1786.336 mm	1224.296 mm	-562.040 mm	-11.714 /mm	-12.259 /mm	562.040 mm	withheld
Roll Front-View Swing Arm	rear	1786.336 mm	1224.296 mm	-562.040 mm	-120.970 /deg	-121.625 /deg	562.040 mm	withheld
Bump Front-View IC z	rear	-57.864 mm	-91.181 mm	-33.316 mm	-3.964 /mm	-1.564 /mm	137.147 mm	withheld
Roll Front-View IC z	rear	-57.864 mm	-91.181 mm	-33.316 mm	-40.554 /deg	-14.717 /deg	73.752 mm	withheld
Roll RC z-Migration	rear	-22.682 mm	-46.976 mm	-24.293 mm	0.000 /deg	0.000 /deg	37.144 mm	withheld
Bump RC Height	rear	-18.080 mm	-43.564 mm	-25.484 mm	-1.751 /mm	-1.576 /mm	31.743 mm	withheld
Bump RC z-Migration	rear	-22.682 mm	-46.976 mm	-24.293 mm	-0.751 /mm	-0.576 /mm	30.553 mm	withheld
Roll RC Height	rear	-18.080 mm	-43.564 mm	-25.484 mm	-18.505 /deg	-16.711 /deg	28.052 mm	withheld
Bump RC Migration	rear	-0.000 mm	0.000 mm	0.000 mm	-0.751 /mm	-0.576 /mm	7.964 mm	withheld
Roll RC Migration	rear	-0.000 mm	-0.000 mm	0.000 mm	-7.925 /deg	-6.130 /deg	2.831 mm	withheld
Bump RC z-Migration	front	-31.724 mm	-31.724 mm	0.000 mm	-0.615 /mm	-0.615 /mm	0.000 mm	withheld
Roll RC z-Migration	front	-31.724 mm	-31.724 mm	0.000 mm	0.000 /deg	0.000 /deg	0.000 mm	withheld
Bump Front-View IC z	front	-102.505 mm	-102.505 mm	0.000 mm	-4.473 /mm	-4.473 /mm	0.000 mm	withheld

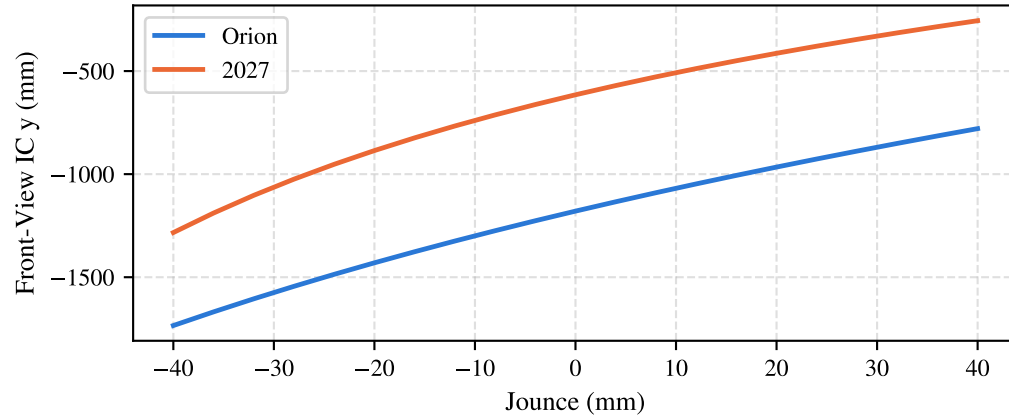
Summary - design position and working-range gradient (3/3)

Values at zero are read from the sweep, not extrapolated. Slope is the least-squares gradient over the full swept range. 'x tol' is how many engineering tolerances the peak change spans.

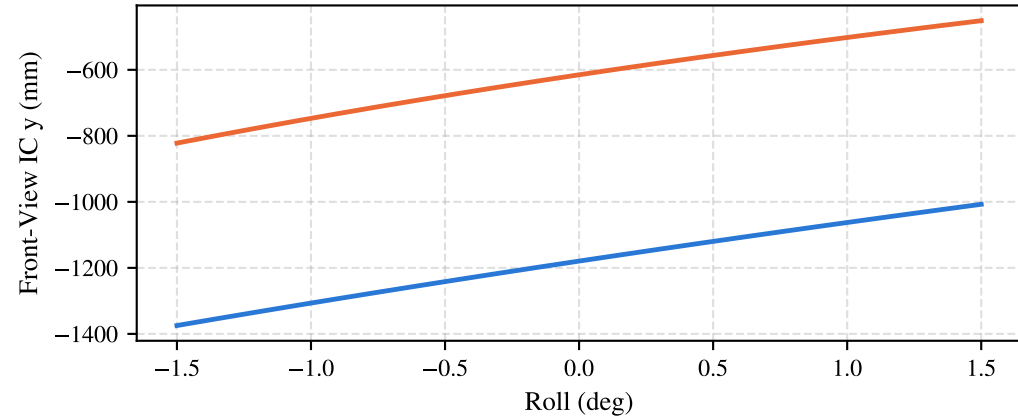
Curve	Axle	Orion @0	2027 @0	delta @0	Orion slope	2027 slope	peak delta	x tol
Bump Front-View Swing Arm	front	2190.086 mm	2190.086 mm	0.000 mm	-8.577 /mm	-8.577 /mm	0.000 mm	withheld
Bump RC Height	front	-27.122 mm	-27.122 mm	0.000 mm	-1.615 /mm	-1.615 /mm	0.000 mm	withheld
Bump RC Migration	front	0.000 mm	0.000 mm	0.000 mm	-0.615 /mm	-0.615 /mm	0.000 mm	withheld
Roll Front-View IC z	front	-102.505 mm	-102.505 mm	0.000 mm	-46.997 /deg	-46.997 /deg	0.000 mm	withheld
Roll Front-View Swing Arm	front	2190.086 mm	2190.086 mm	0.000 mm	-90.332 /deg	-90.332 /deg	0.000 mm	withheld
Roll RC Height	front	-27.122 mm	-27.122 mm	0.000 mm	-17.067 /deg	-17.067 /deg	0.000 mm	withheld
Roll RC Migration	front	0.000 mm	0.000 mm	0.000 mm	-6.487 /deg	-6.487 /deg	0.000 mm	withheld

Headline - changes beyond engineering tolerance

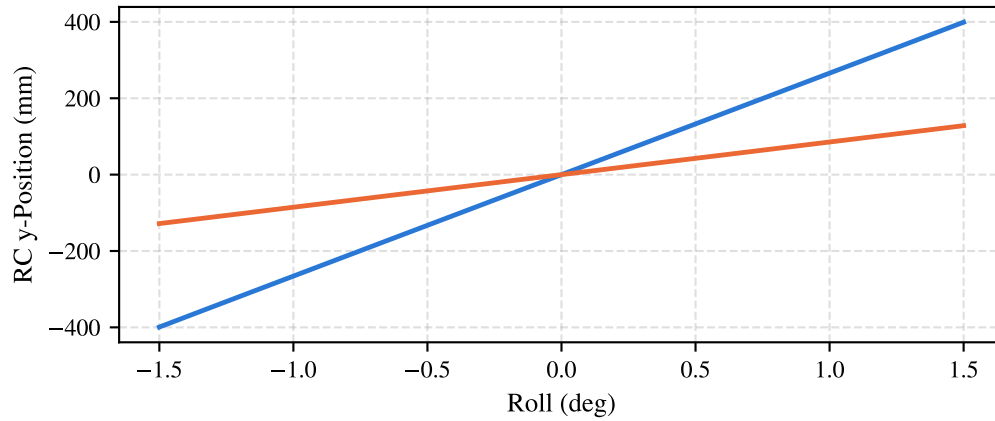
Rear - Bump Front-View IC y



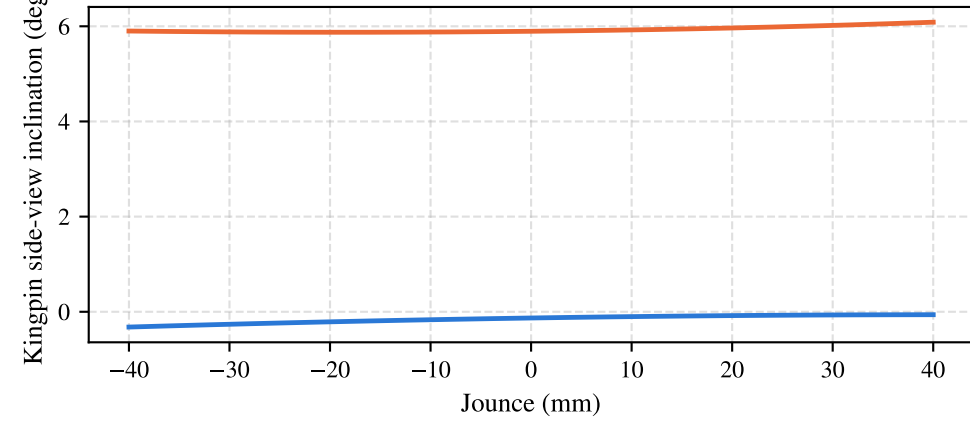
Rear - Roll Front-View IC y



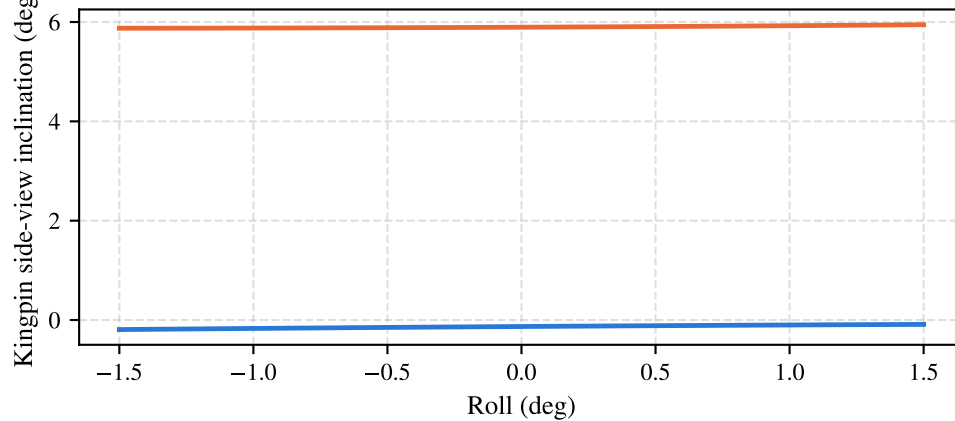
Rear - Roll RC y-Migration



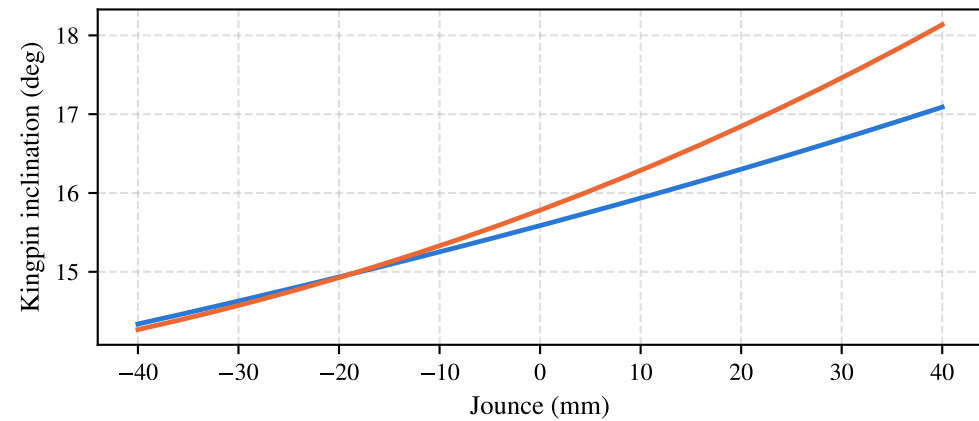
Rear - Bump Kingpin side-view inclination



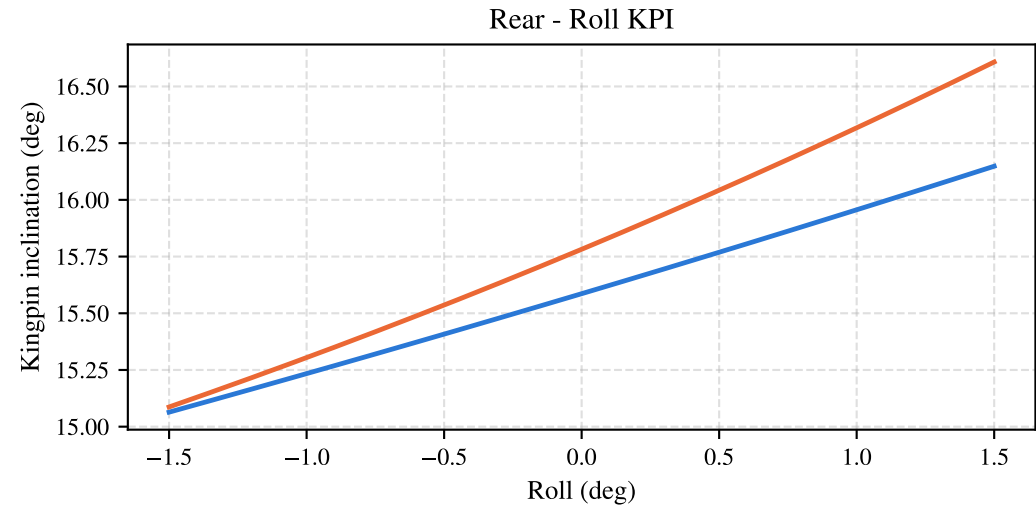
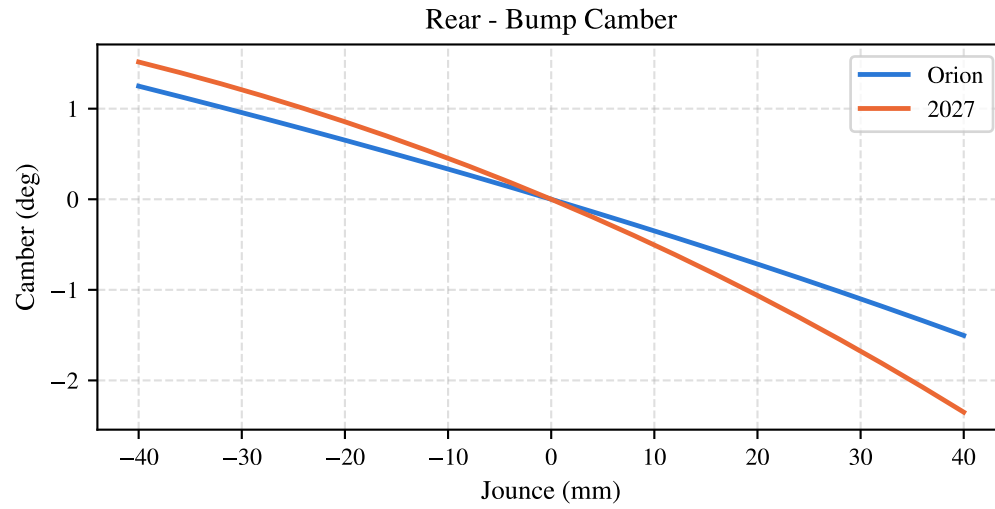
Rear - Roll Kingpin side-view inclination



Rear - Bump KPI

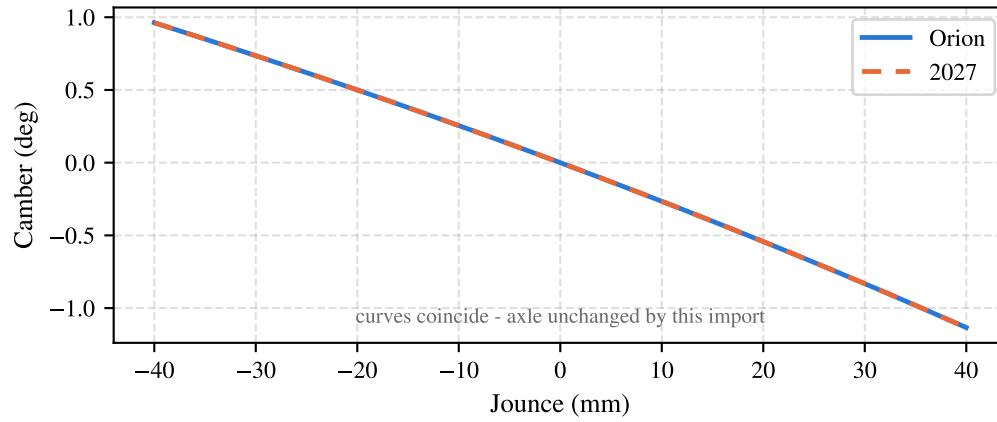


Headline - changes beyond engineering tolerance (continued)

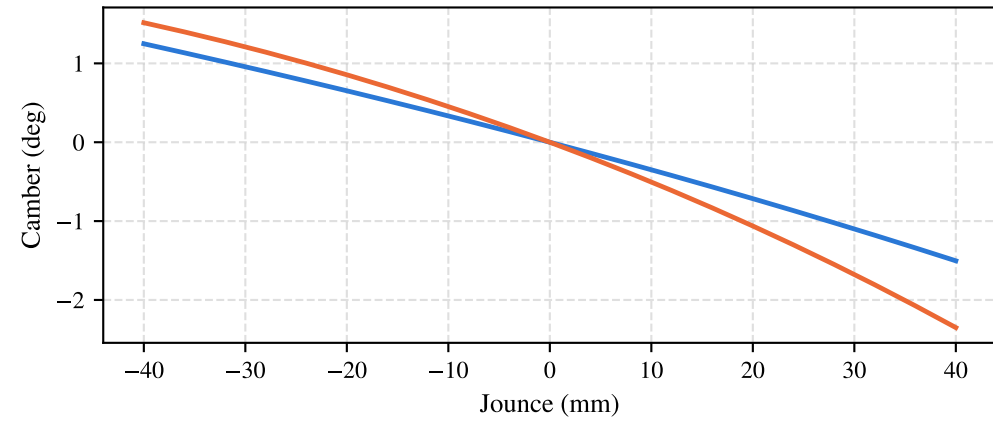


Appendix - full curve deck (1/7)

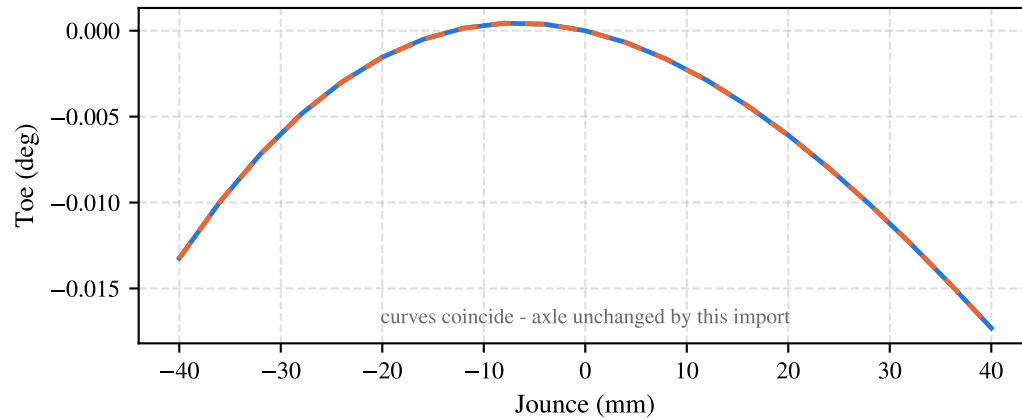
Front - Bump Camber



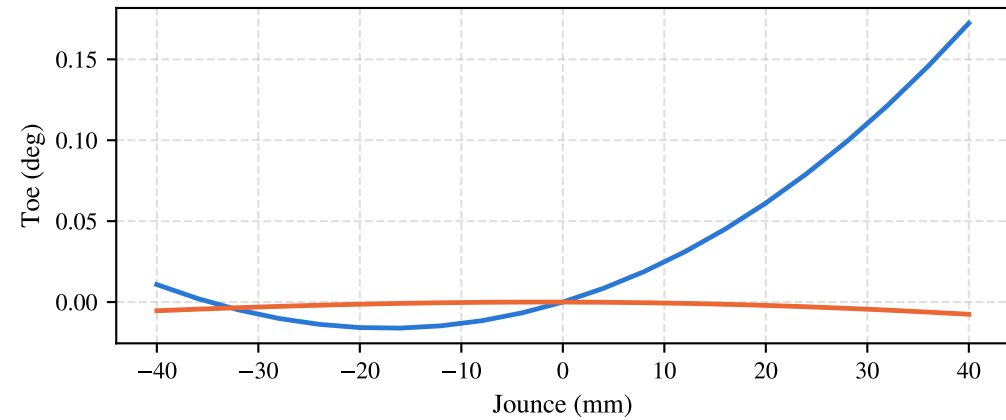
Rear - Bump Camber



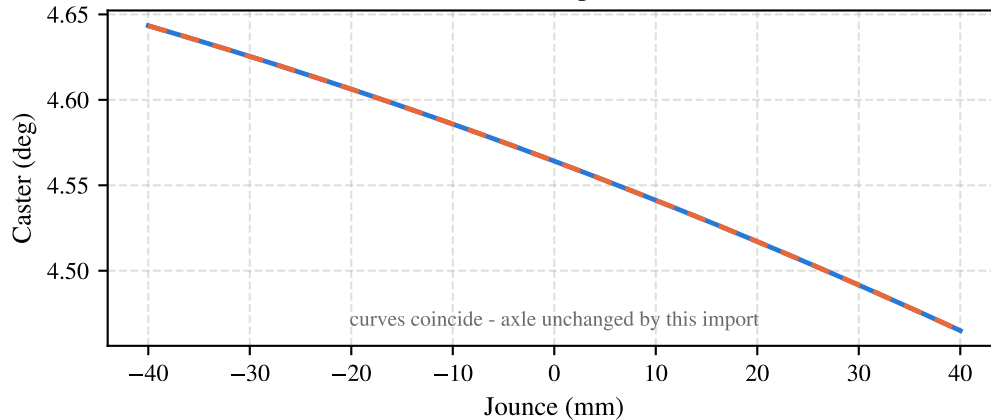
Front - Bump Toe



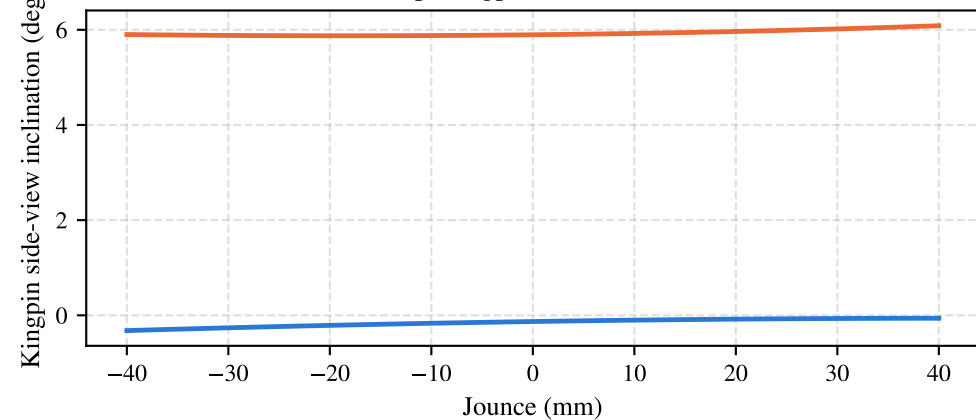
Rear - Bump Toe



Front - Bump Caster

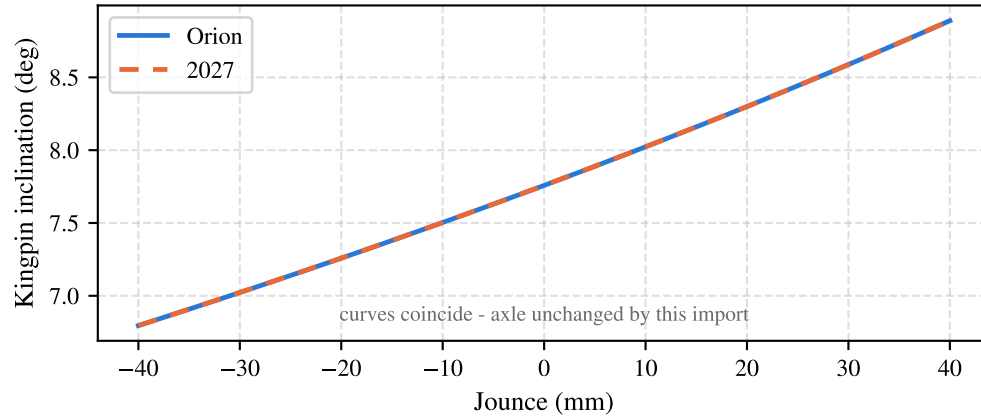


Rear - Bump Kingpin side-view inclination

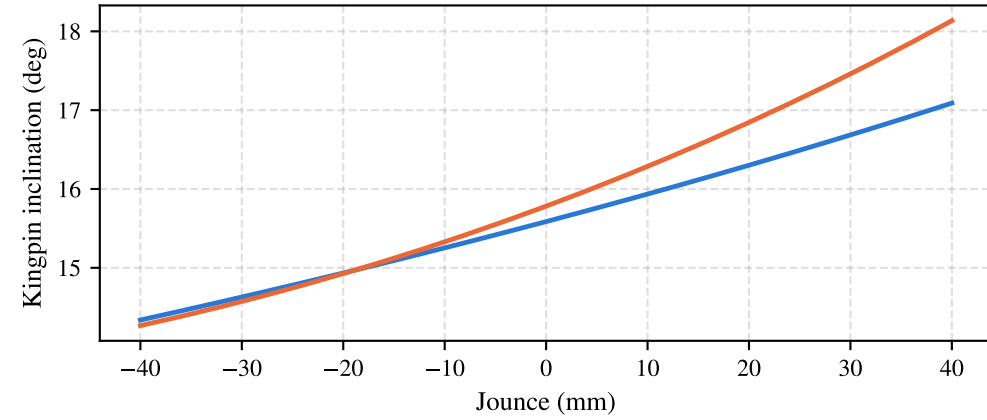


Appendix - full curve deck (2/7)

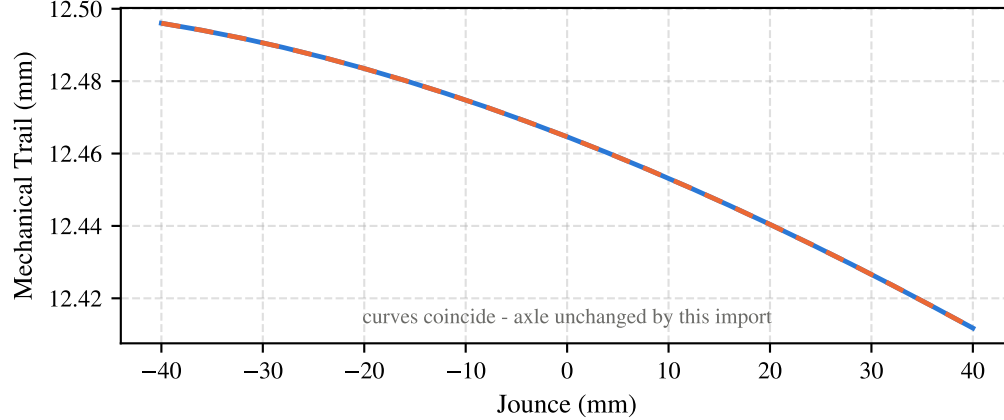
Front - Bump KPI



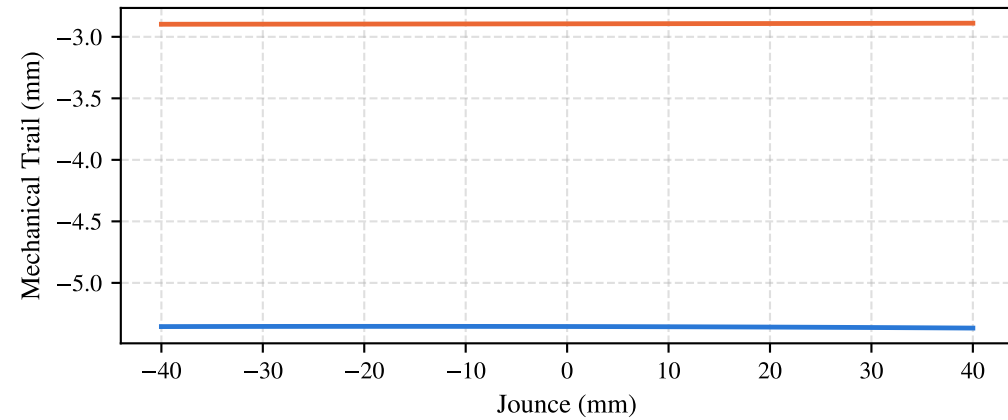
Rear - Bump KPI



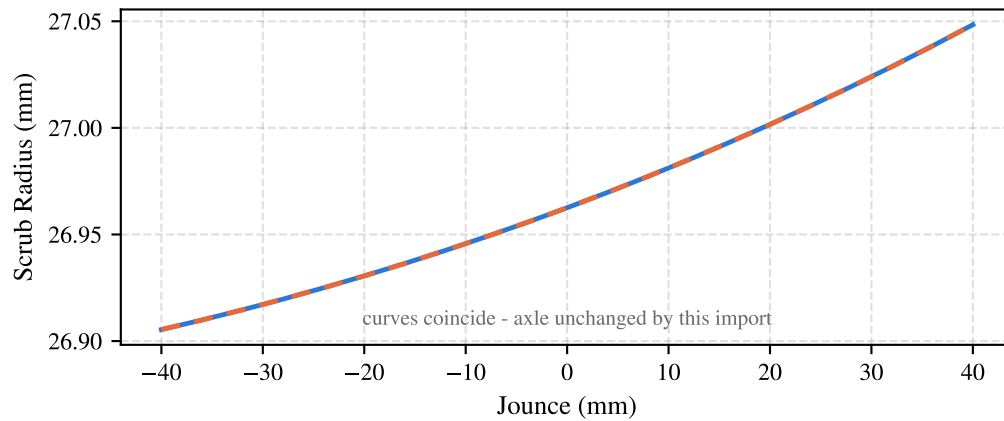
Front - Bump Mechanical Trail



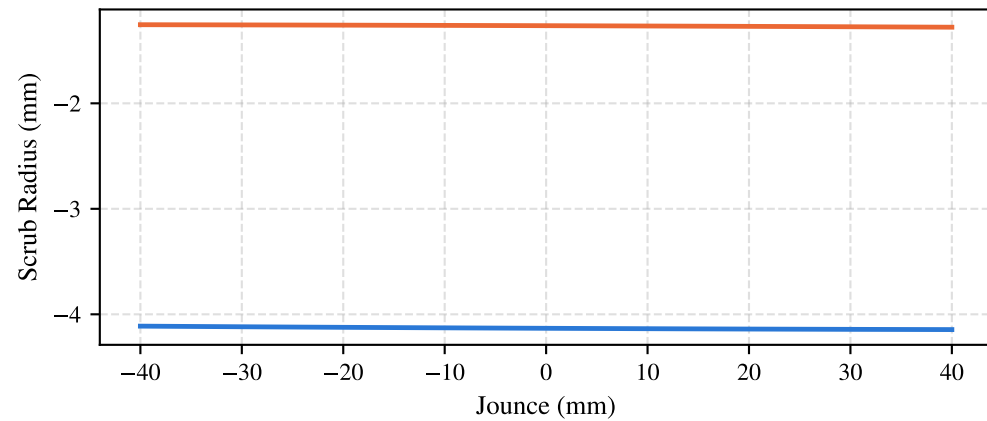
Rear - Bump Mechanical Trail



Front - Bump Scrub Radius

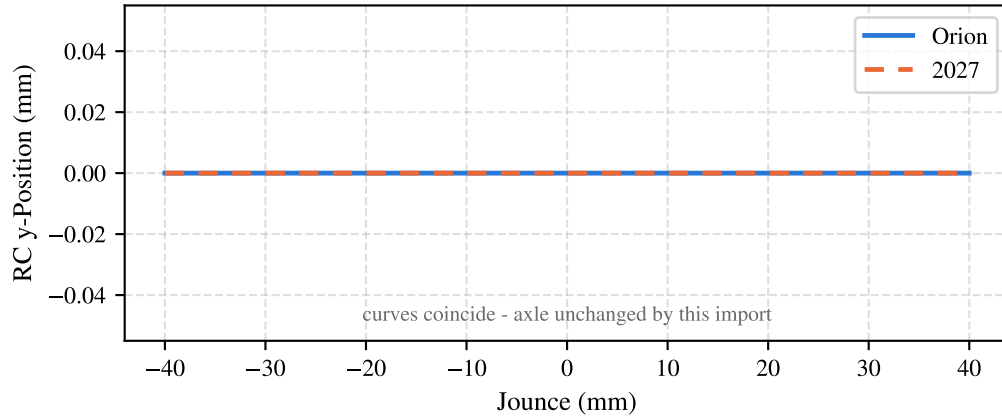


Rear - Bump Scrub Radius

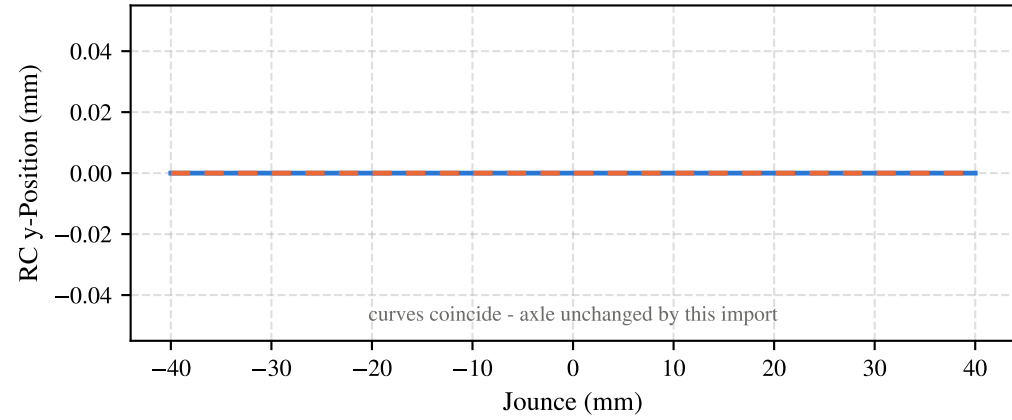


Appendix - full curve deck (3/7)

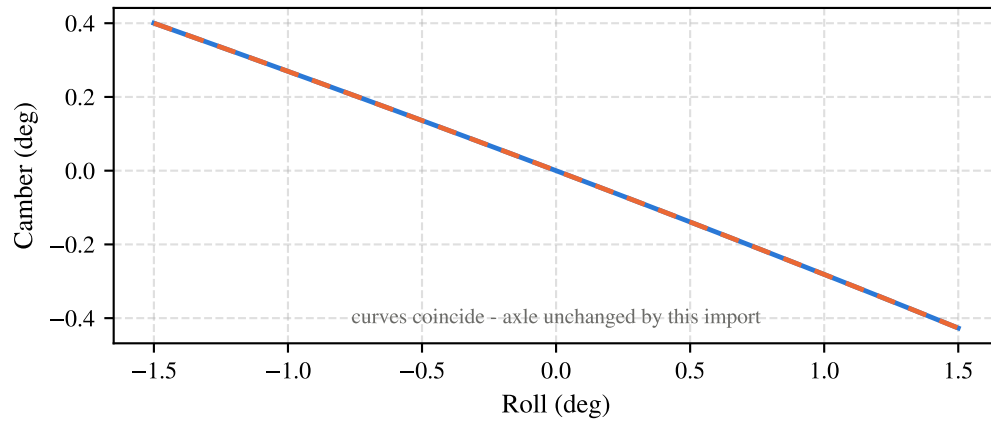
Front - Bump RC y-Migration



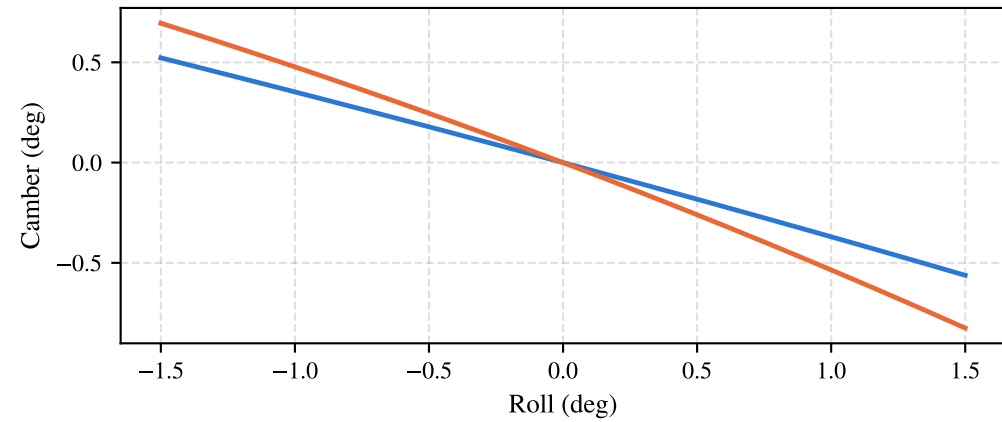
Rear - Bump RC y-Migration



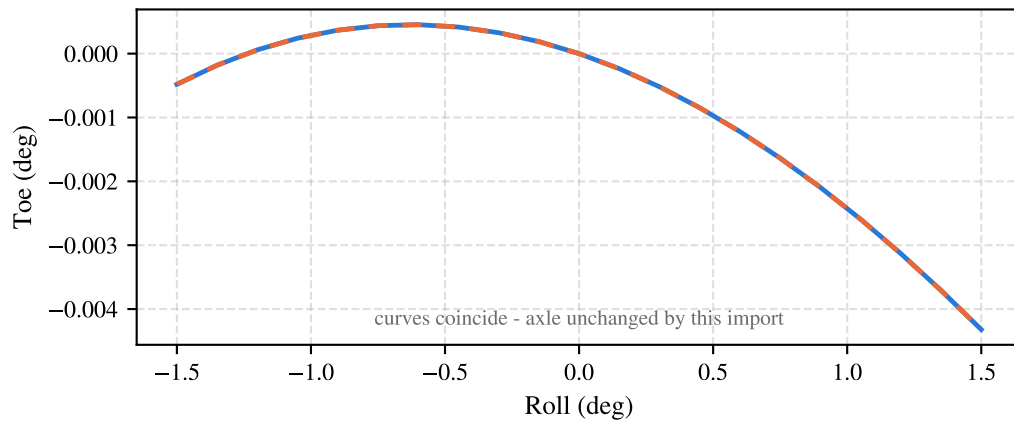
Front - Roll Camber



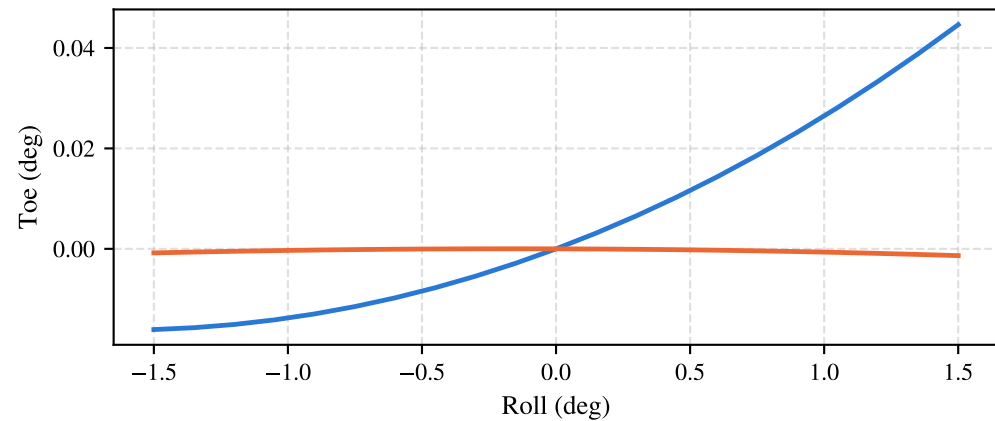
Rear - Roll Camber



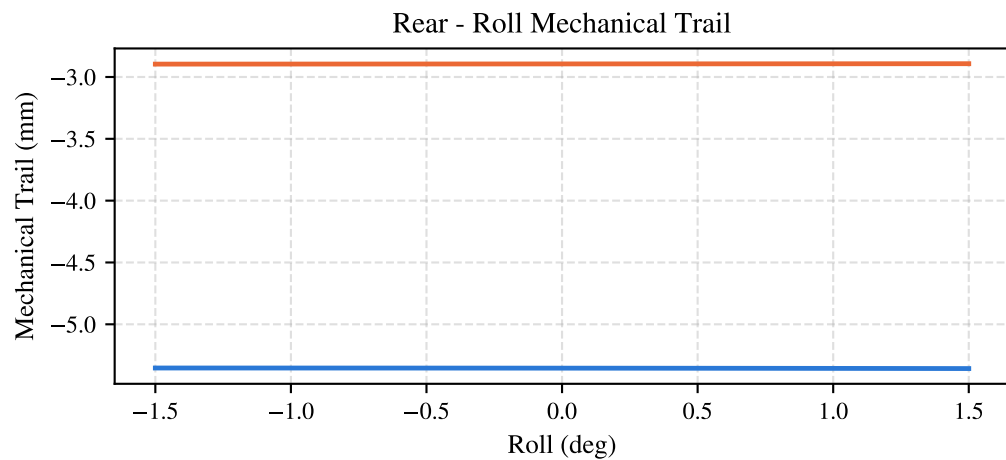
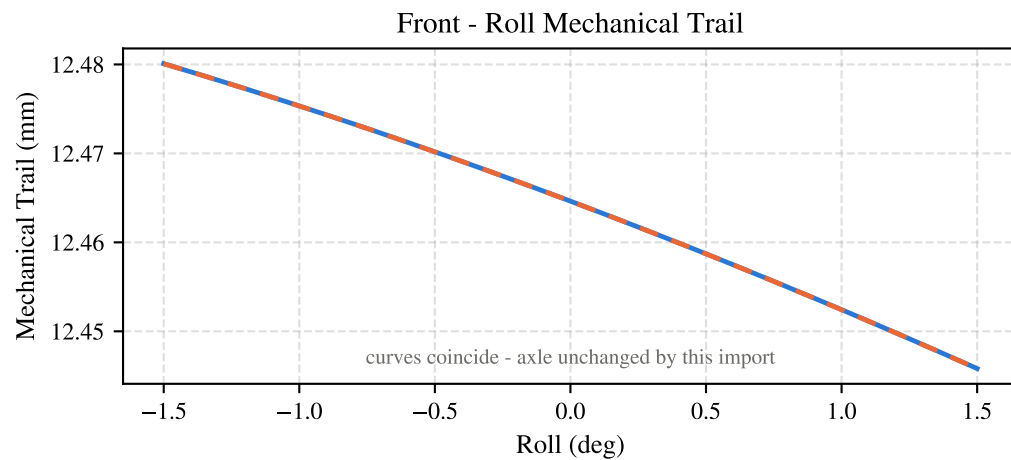
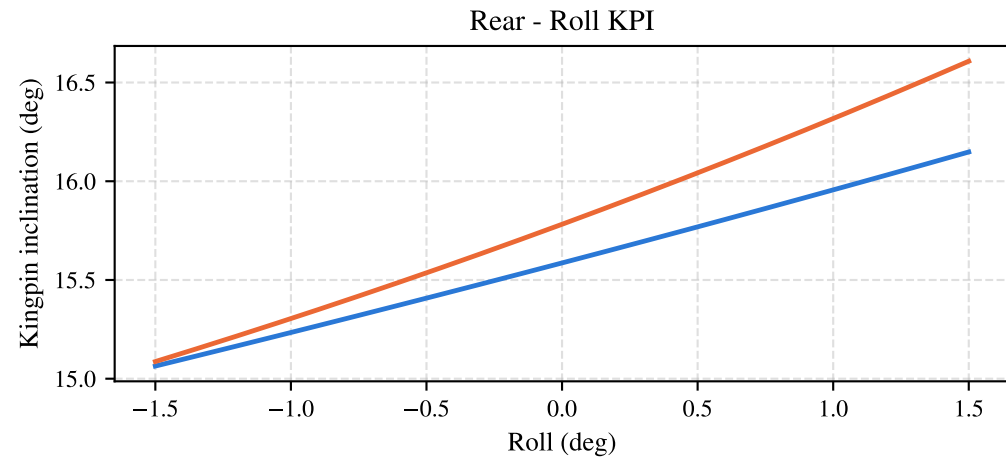
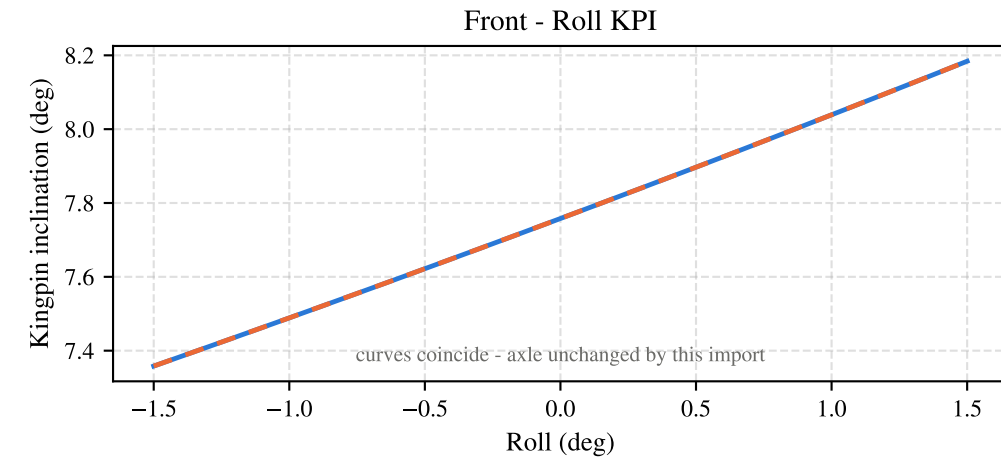
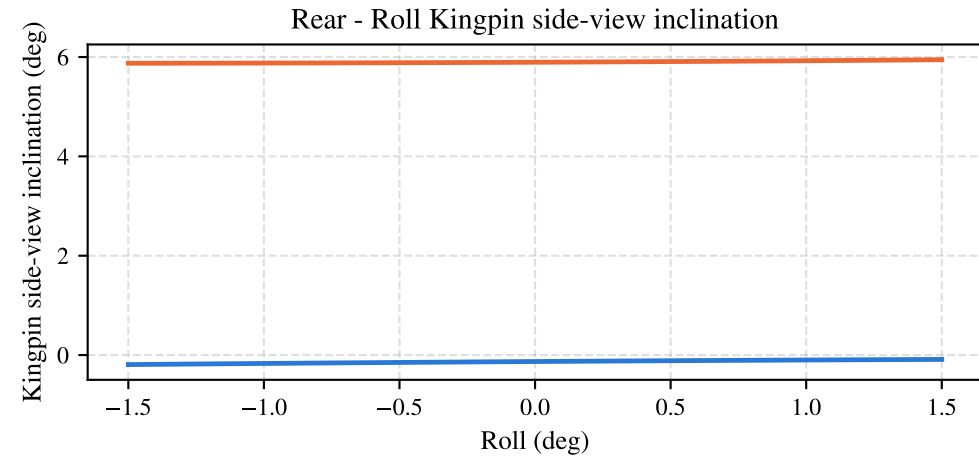
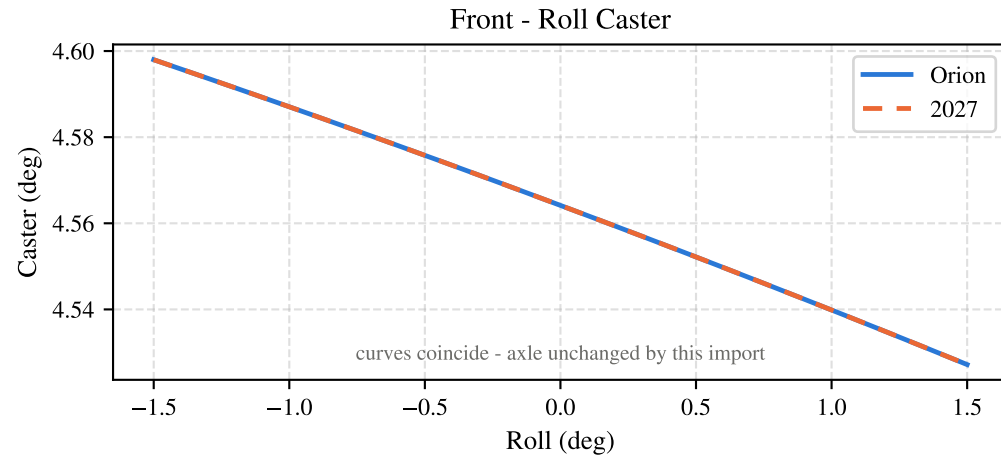
Front - Roll Toe



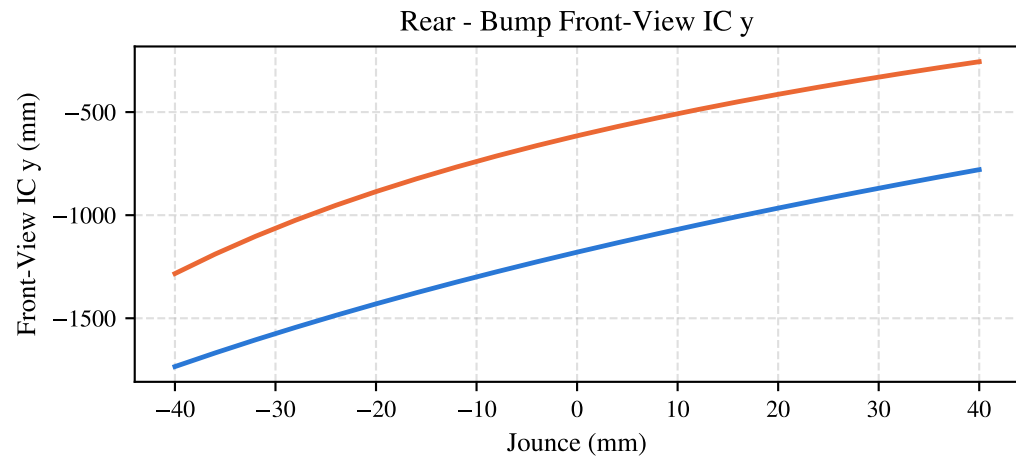
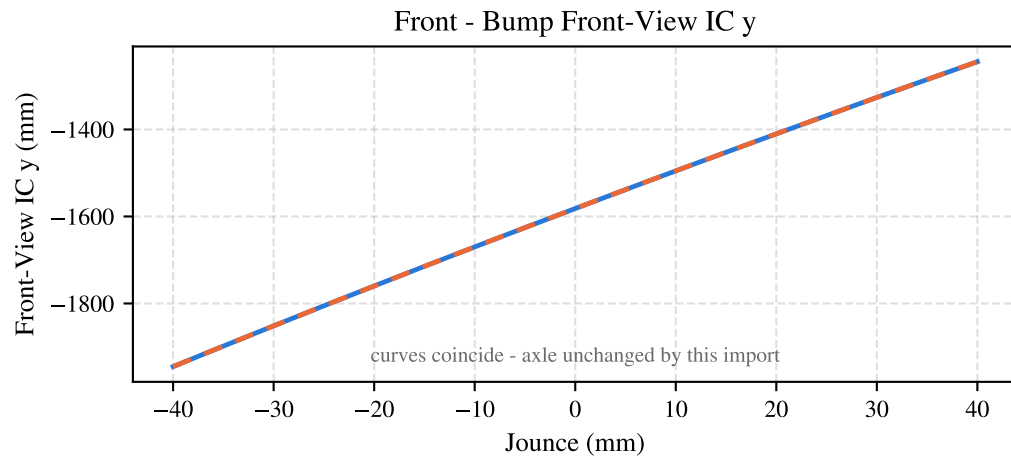
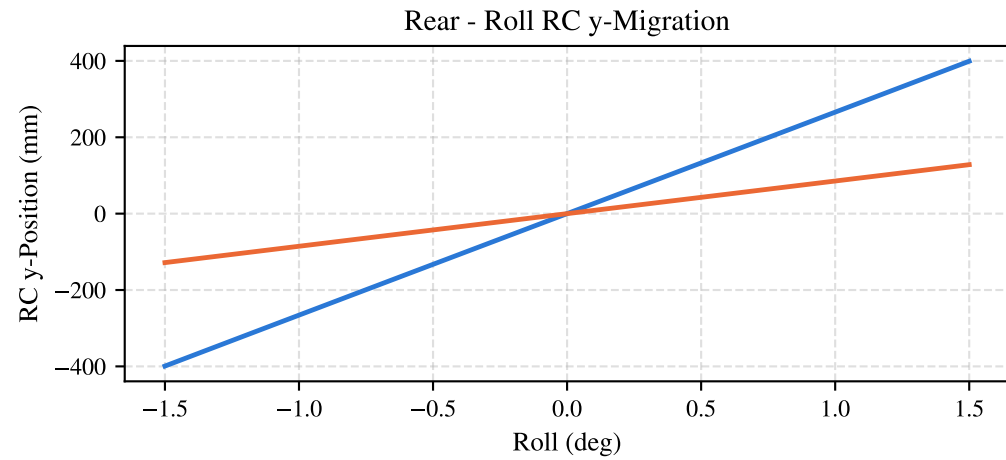
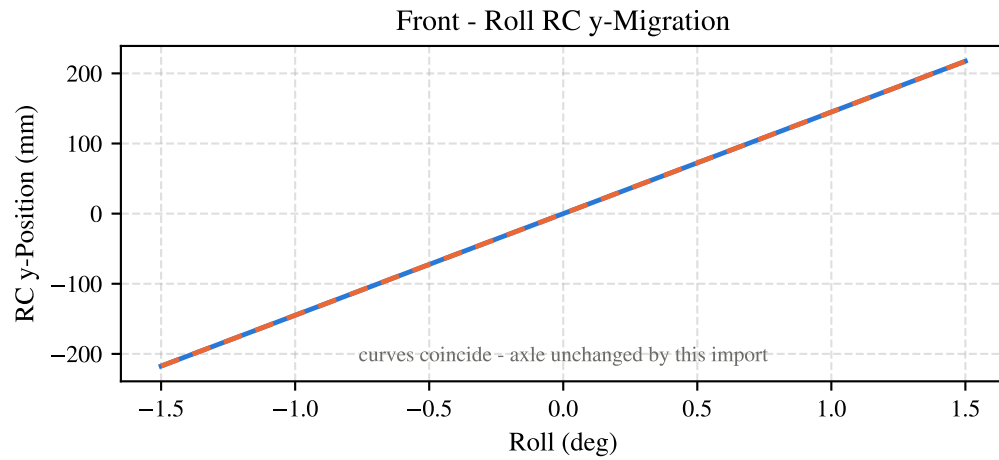
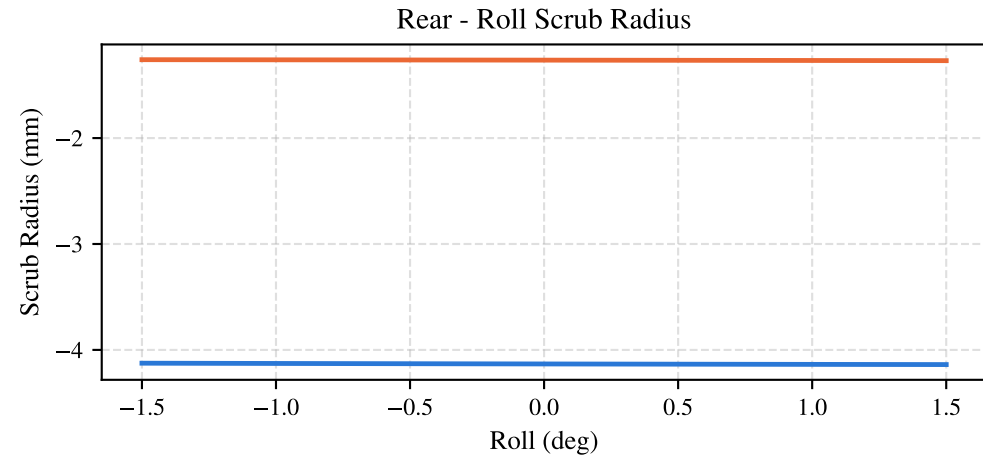
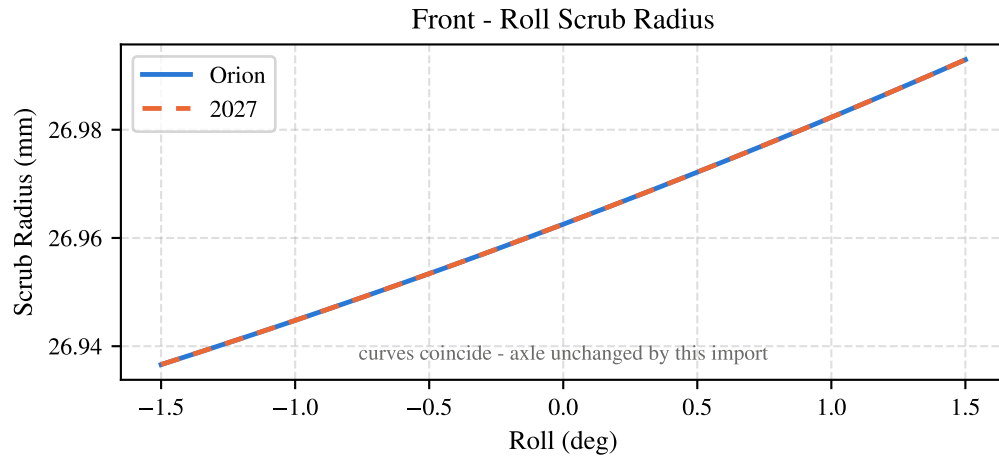
Rear - Roll Toe



Appendix - full curve deck (4/7)

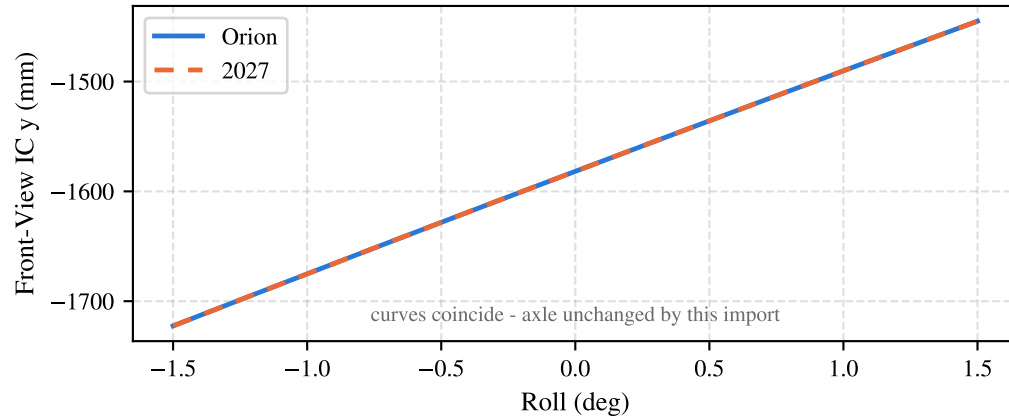


Appendix - full curve deck (5/7)

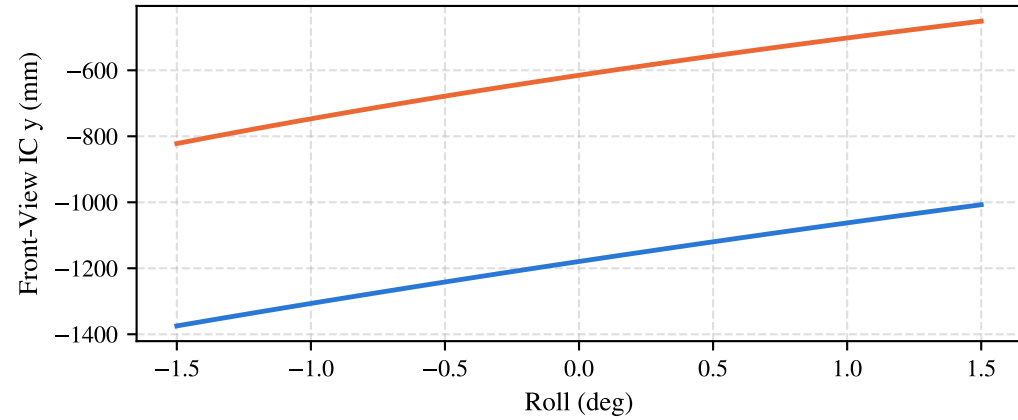


Appendix - full curve deck (6/7)

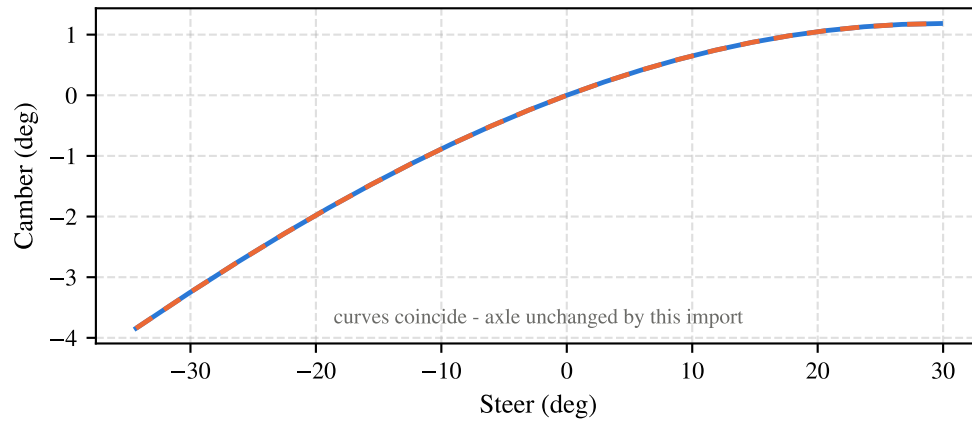
Front - Roll Front-View IC y



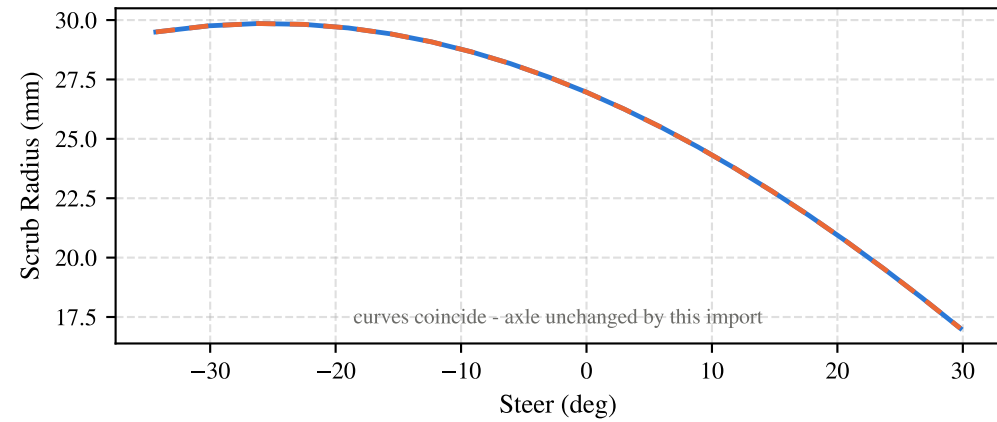
Rear - Roll Front-View IC y



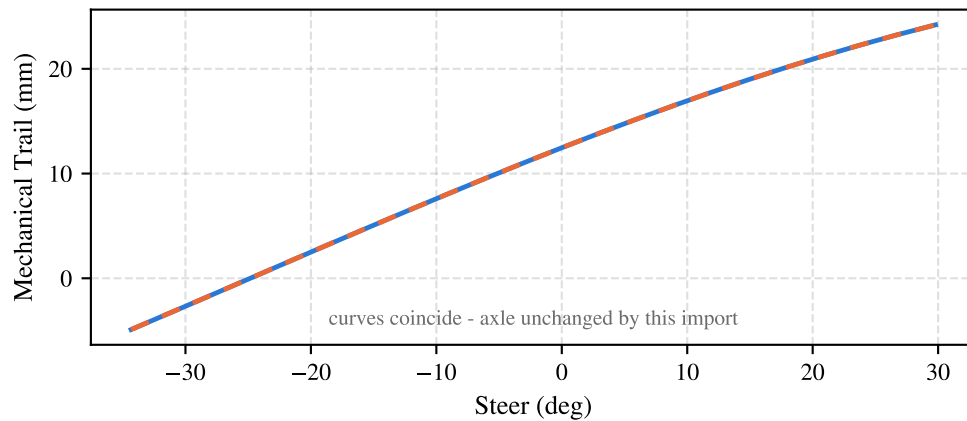
Front - Steer Camber



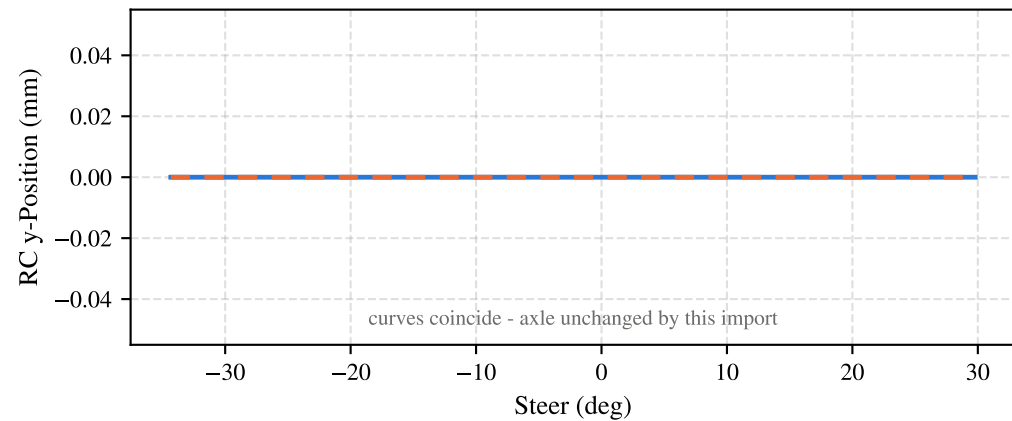
Front - Steer Scrub Radius



Front - Steer Mechanical Trail

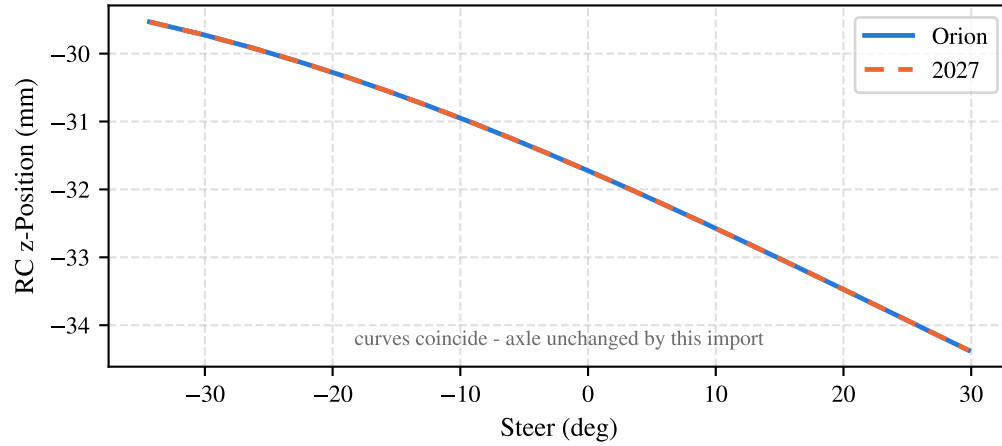


Front - Steer RC y-Migration

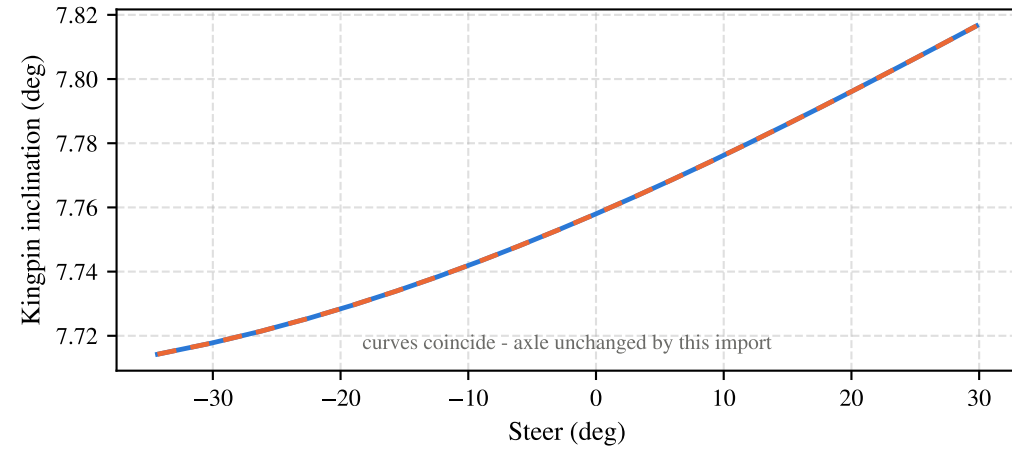


Appendix - full curve deck (7/7)

Front - Steer RC z-Migration



Front - Steer KPI



Front - Steer Caster

